International Journal of Medical Informatics 195 (2025) 105799

Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf

Automatic (near-) duplicate content document detection in a cancer registry

Tapio Niemi *, Jean Pierre Ghobril, Gautier Defossez, Simon Germann, Eloïse Martin, Jean-Luc Bulliard

Centre for Primary Care and Public Health (Unisanté), University of Lausanne, Lausanne, Switzerland

ARTICLE INFO

Keywords:
(near-) duplicate documents
Locality sensitive hashing
Cancer registry
Smith-Waterman

ABSTRACT

Background: Duplicate and near-duplicate medical documents are problematic in document management, clinical use, and medical research. In this study, we focus on multisourced medical documents in the context of a population-based cancer registry in Switzerland. Although the data collection process is well-regulated, the volume of transmitted documents steadily increases and the presence of full or near-duplicates slows down and complicates document processing. Identifying near-duplicates is particularly challenging because the large number of documents makes pairwise comparison non-feasible.

Methods: We implemented a system based on both normal hash functions, Simhash (Locality Sensitive Hashing), and Smith-Waterman text alignment similarity. Simhash offers good performance and confirming its results by the Smith-Waterman algorithm with a selected similarity threshold reduces the false positive rate to near zero without lowering sensitivity. Extracted differences in near-duplicate content documents are shown by highlighting differences in original PDF documents.

We validated the method using 3042 manually verified document pairs containing 1252 full-duplicate and 398 near-duplicate pairs. The area under the curve (AUC) was 0.96, sensitivity 0.92, specificity 1.00, PPV 1.00, and NPV 0.91. For the same size simulated data, corresponding values were 0.86, 0.72, 1.00, 1.00, and 0.77, respectively.

Results: We applied the method against 224,398 medical documents in the cancer registry. We found 5.5% of duplicates on the text level, and 0.17–0.24% near-duplicates depending on the used parameters and threshold values. Most near-duplicates related to the same patient and originated from the same transmitter. Manual evaluation showed that only 2% of differences were in medical contents and 83% in administrative data (21% in patient, 11% in doctor, and 51% in other administrative data). Many near-duplicates looked strikingly similar from a human perspective.

Conclusions: We demonstrated that our method can efficiently find all full-duplicates and most near-duplicates in a large set of multisourced medical documents. Potential ways to further improve this method are discussed. The method can be applied to documents in all domains.

1. Introduction

Oncological diseases are a major public health issue worldwide and their frequencies are monitored in many countries through population-based registries. Data stored in these registries are utilised in epidemiological and clinical studies such as quality of care, treatment delay and compliance. In Switzerland, where regional cancer registries operate since the 1970 s, reporting of cancer cases has been mandatory since 2020 for all institutions and treating physicians [1]. This legal change led to a huge increase in the number of transmitted medical documents to cancer registries from healthcare providers, including medical cabinets, private and public hospitals, and pathology laboratories. Data sent on new cancer cases include both structured data and textual documents.

Several studies have shown that medical documents contain a lot of duplicated or redundant text, mainly due to manually copied text or automatically generated updated documents [2,3]. Further, the same documents are sometimes sent by various transmitters with possibly

* Corresponding author.
E-mail address: tapio.niemi@unisante.ch (T. Niemi).

https://doi.org/10.1016/j.ijmedinf.2025.105799
Received 16 June 2023; Received in revised form 14 January 2025; Accepted 16 January 2025
Available online 18 January 2025

different formatting. This creates duplicate or near-duplicate documents which can cause duplicated counts in statistics, lower performance of AI and machine learning models [4], increased need of storage space, and unnecessary technical complexity in document management. From a non-data science perspective, it causes unnecessary workload for medical professionals using these documents (e.g., physicians and medical coders in hospitals and cancer registries), thus loss of efficiency.

Finding near-duplicate documents in large document collections is challenging since brute force-based pairwise comparison is usually too slow because of its quadratic time complexity. The search for an efficient solution to this problem has attracted interest over the last 25 years mostly because of its importance for Internet search engines: the result of a search should not contain (near-) duplicate web pages (e.g., [5,6]). Another significant application has been plagiarism detection ([7,8]).

Most of the existing methods are based on computing document 'fingerprints'. A fingerprint can be only a couple of bytes long making storage and comparison more efficient than operations with the original documents. A traditional hash fingerprints are the same if the documents are exactly the same (efficient for full-duplicates). However, traditional hash fingerprints do not allow a similarity estimation since only a tiny difference leads to completely different fingerprints. Therefore, Locality Sensitive Hashing (LSH) methods, have been developed for near-duplicate detection. Unlike other hashing methods, LSH creates specific fingerprints that allow computing differences between documents from the fingerprints without requiring access to original documents. The two most used LSH methods are MinHash [9] and Simhash [10]. Both methods originate from detecting near-duplicate web pages and yield sufficient performance even for billions of documents ([11,6]).

Other methods are based on pairwise similarity comparison. They apply metrics such as the Levenshtein distance [12], Jaccard's similarity [13], and a text alignment algorithm by Smith and Waterman [14] (S-W). The S-W algorithm was developed to identify common molecular subsequences for biogenetic research but later used for finding similar parts in text documents. The algorithm finds sequences of similar characters or words and returns a similarity score between the documents.

Research work in near-duplicate document detection has focused on IT domain and, to the best of our knowledge, there has been only one study in medical field by Gabriel et al. [4] (a method based on the MinHash for near-duplicate clinical notes).

Our aim was to develop an efficient method for detecting similar documents in a large collection of medical documents sent to the Vaud Cancer Registry in Switzerland. However, the requirements were common to other medical domains: 1) testing whether a new document has near-duplicates among the existing ones should be efficient; 2) fingerprints should be computed independently for each document without prior knowledge of the global vocabulary; 3) the method should be sensitive to small details such as special terms, codes, and numbers; finally, 4) differences in near-duplicate documents should be highlighted for the reader. This all means that methods developed for other domains do not necessarily work well.

Based on the literature and our preliminary testing, we developed a method combining the Simhash and S-W algorithms. We validated it by using a human verified data set and applied it to routine medical documents of the cancer registry.

2. Material and methods

2.1. Data

We used two data sets for validation: 1) a manually validated set of 3042 document pairs (Data Set 1), 2) a simulated data set of 3042 documents based on making tiny modifications in real medical documents (Data Set 2). For final testing, we used a large production data set of 224,398 medical documentscollected between 2020 and 2022 (Data Set 3). (Detailed descriptions in Appendix A).

Fig. 1 illustrates the difference between computer and human detection of full-duplicates in Data set 1. Sensitivity in human detection was 1.0 and precision 0.74, indicating that all duplicate pairs were found but some found pairs were not full-duplicates.

2.2. Method

The Simhash method computes a numeric fingerprint for each document in a way that fingerprints of similar documents are closer to each other than to those of different documents. This makes it possible to detect near-duplicates by comparing their fingerprints. Computing a fingerprint depends only on the document at hand, not the whole documents collection. It is efficient to detect whether a new document has duplicates by finding other documents whose fingerprints differ only in k-bits. Simhash requires only one relatively small numeric fingerprint to represent the document [6].

Simhash, as hash methods in general, has a time complexity of O(n), where n is the number of documents. However, the size of the document impacts the computing time, too. The Simhash method also needs an index structure for finding efficiently similar fingerprints. While traditional hash-based methods give exact and correct results, Simhash is an estimation algorithm. It rarely gives false positive results but can give a small proportion of false negatives (4.5 and 14.4 % in our validation data sets), i.e. it does not detect that two documents with only small differences are near-duplicates. This can be improved by computing two or more Simhash fingerprints with slightly different parameters.

Near-duplicate documents detected by Simhash can have quite large differences in their similarities. Because we are interested in 'almost identical' near-duplicates, we applied the S-W algorithm [14] for computing exact differences between each near-duplicate document. This reduces false positives and is relatively fast since we only need to compare candidate pairs detected by Simhash. Simhash estimates the cosine similarity, while the S-W similarity is an edit distance. Unlike many other similarity metrics, the S-W algorithm performs local sequence alignment by finding the most similar subsequence(s) within two large sequences without penalizing the gaps disproportionally. This is because in DNA sequence alignment both small changes (point substitutions, typos in text) as well as large changes (translocations, insertions, deletions) count. Therefore, S-W similarities correspond well to a human observer's evaluations in Data Set 1 (Fig. 5, Table 1). As a baseline control measure, we used a metric based on Hamming [15] distance, that is, the number of positions in two strings in which the characters are different. Altogether, we defined three normalised similarity measures: Simhash, Hamming, and S-W similarity (Appendix B).

Finally, we manually examining differences in near-duplicates and classified them into four groups based on types of differences between the documents. The groups were: 1) medical data, 2) administrative data in three subgroups (patient data, doctor or clinic data, and other administrative data), 3) no visible difference, and 4) not a near-duplicate. If a near-duplicate pair contained more than one difference type, the first classification criterion was medical information followed by administrative data.

2.3. Algorithm for near-duplicate detection

The developed method (Fig. 2) is based on PDF text extraction, computing multiple hash fingerprints (normal hash, Simhash), pairwise comparison of near-duplicate candidates by using the S-W algorithm, and data extracted from the standard document processing in the Vaud Cancer Registry. Additionally, our method can also extract other information stored in PDF files such as highlighted text in the documents. Detailed descriptions of our method as well as Simhash algorithm are given in Appendix C.

We defined parameters for the Simhash method as follows. We used $k = 3$ (the maximum number of different bits in fingerprints) [6] for Simhash and 0.94 as the threshold value for S-W (the optimal threshold

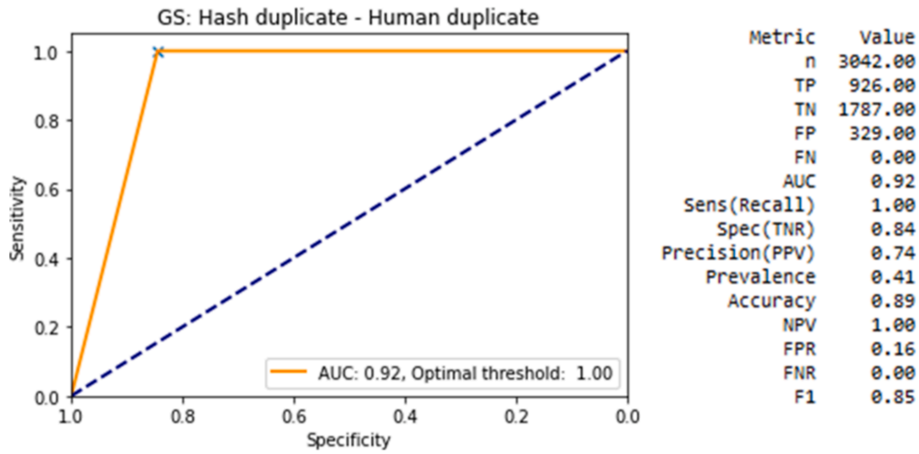

Fig. 1. Comparison of hash function duplicates as a gold standard to human detected duplicates in Data set 1. (The orange line represents sensitivity as a function of specificity. The diagonal dashed blue line indicates performance of a random guess model. The area under the curve (AUC) measures the goodness of the model (1.0 for a perfect model, 0.5 for a random guess). Optimal threshold is based on Youden's index). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Comparison of Simhash, S-W, Hamming similarity metrics, and our method combining Simhash and S-W against the human validated gold standard (Data Set 1) and the simulated data set (Data Set 2). We used 0.35 as a threshold value for the Simhash in Data Set 2 for improving sensitivity (optimal based on Youden's index were 0.58). Simhash parameters: width = 6 words, overlap = 3.

Metric	Simhash		S-W		Hamming		Simhash + S-W S-W for Simhash positives		Simhash + S-W combined results	
	Data Set 1	Data Set 2	Data Set 1	Data Set 2	Data Set 1	Data Set 2	Data Set 1	Data Set 2	Data Set 1	Data Set 2
n	3042	3042	3042	3042	3042	3042	1518	1546	3042	3042
TP	1512	1099	1607	1537	1227	1518	1511	1099	1511	1099
TN	1386	1058	1388	1504	1378	1465	2	446	1388	1504
FP	6	447	4	1	14	40	4	1	4	1
FN	138	438	43	0	423	19	1	0	139	438
Sensitivity	0.92	0.72	0.97	1.00	0.74	0.99	0.67	1.00	0.92	0.72
Specificity	1.00	0.70	1.00	1.00	0.99	0.97	1.00	1.00	1.00	1.00
PPV	1.00	0.71	1.00	1.00	0.99	0.97	0.33	1.00	1.00	1.00
Accuracy	0.95	0.71	0.98	1.00	0.86	0.98	1.00	1.00	0.95	0.86
NPV	0.91	0.71	0.97	1.00	0.77	0.99	1.00	1.00	0.91	0.77
FPR	0.00	0.30	0.00	0.00	0.01	0.03	0.67	0.00	0.00	0.00
FNR	0.08	0.28	0.03	0.00	0.26	0.01	0.67	0.00	0.08	0.28
F1	0.96	0.71	0.98	1.00	0.85	0.98	0.00	1.00	0.96	0.84

Sensitivity (recall) = TP/(TP + FN), specificity = TN/(TN + FP), F1 = 2 x (precision x sensitivity) / (precision + sensitivity), PPV (positive predictive value or precision) = TP/(TP + FP),NPV (negative predictive value) = TN/(TN + FN), FPR (false-positive rate) = FP/(FP + TN), FNR (false-negative rate) = FN/(FN + TP).

obtained from Data set 1). We used Data Sets 1 and 2 for selecting optimal shingle length and overlap parameters for Simhash. The shingle width and overlap together impact performance of the Simhash method. The overlap must be greater than zero to account for the order of shingles (or words in the text). However, in absence of a generic heuristic to define these parameters, we used an empirical method to select the most suitable values. Fig. 3 illustrates the goodness of different combinations in Data Set 1. The shingle width 6 and overlap 3 gave the best sensitivity and specificity.

Although near-duplicates can be detected with high precision, the automatic method cannot identify which document in the near-duplicates is the 'correct' one. Therefore, the final step in our method is visualising differences in near duplicates for medical coders. First, the differences in the original PDF text are detected by using Python's difflib module and then their locations in PDFs are found. Finally, both PDFs with highlighted differences are shown side by side as illustrated in Fig. 4.

2.4. Validation

We used two data sets for comparing Simhash and S-W similarities against the human validated gold standard in Data Set 1 and simulated data in Data Set 2.

We compared Simhash and S-W similarity measures among document pairs in Data Set 1 (Figure B-1, Appendix B). The comparison highlights that a high Simhash similarity indicates high S-W similarity but document pairs with high S-W similarities can have low Simhash similarities. This means that Simhash has a lower sensitivity than S-W.

Fig. 5 shows Receiver Operating Characteristic (ROC) curves for Simhash, S-W, Hamming and Simhash – S-W combination (our method) similarities against human validated gold standard (Data Set 1). The S-W similarity performs best and clearly outperforms the Hamming similarity used as a baseline control measure. AUC of our method was 0.96.

We computed pairwise similarities among the documents by using both S-W and Hamming similarities to validate our choice of using S-W similarity. The S-W similarity almost perfectly matches the human interpretation, while the strict character wise Hamming similarity clearly differs from the human-defined similarity. These results confirm our assumption of using the S-W similarity.

The results for the simulated Data Set 2 overall corroborated those of Data Set 1 (Table 1). The main difference was that Simhash performed less well in the simulated data. The results of Data Set 2 also showed that the combined Simhash and S-W similarity improved substantially the specificity compared to Simhash alone (1.00 vs 0.70). AUC of our

International Journal of Medical Informatics 195 (2025) 105799

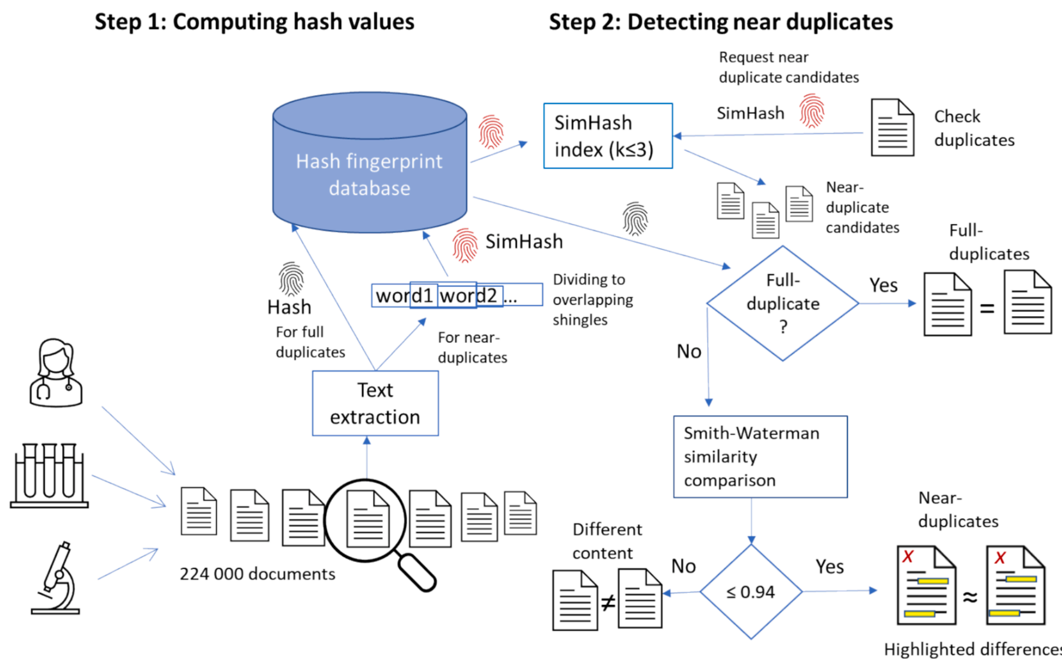

Fig. 2. Graphical illustration of the method. Step 1 contains text extraction, preprocessing, and hash fingerprint computation. Step 2 includes tasks needed in near-duplicate detection.

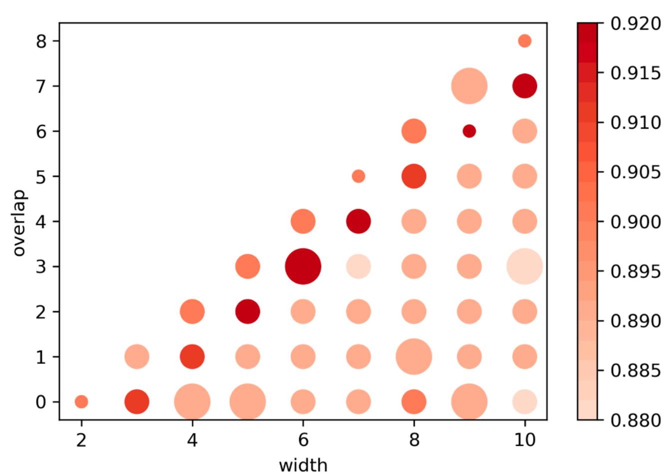

Fig. 3. Sensitivity (colour) and specificity (size) as a function of shingle length and overlap in Data Set 1.

method was 0.86 in Data Set 2.

3. Results

As a final step, we applied our method to a large set of medical documents (Data Set 3). In this case, no gold standard was available and pairwise comparisons using string similarity metrics were unrealistic since it would have required over 21 billion pairs to test. Data Set 3 contained an undetermined number of duplicate documents and also triplicates, quadruplicates, etc. As the unit ‘duplicate pairs’ was not applicable anymore, we expressed the results as the number of documents having at least one (near-) duplicate. This metric gives twice as large values as duplicate pairs does if the data contain only duplicates (no triplicate, quadruplicate, etc.).

The data set contained 224,398 readable medical documents. Of these, 12,256 had at least one full-duplicate on the preprocessed text level. The Simhash algorithm with the predefined parameters width = 6, overlap = 3, and k = 3 (see Methods), returned 12,917 near- or full-duplicate document candidates. After removing full-duplicates, we had 762 near-duplicate candidates. Filtering these using the S-W similarity with the predefined threshold 0.94 (see Methods) gave 540 near-duplicate documents (Table 2).

When we combine the duplicate data with extracted document data, we can classify the duplicates based on further criteria. For example, a large part of both full- and near-duplicates related to the same patient (the same first and last name and date of birth) and had the same transmitter (Table 2).

Our manual classification of near-duplicate document pairs by types of differences indicated that 83 % of duplicates were in administrative data: patient 21 %, doctor or clinic 11 %, and other administrative data (dates of letters, headings, headers/footers, etc.) 51 %. These included small differences or typos in names, addresses, dates, times, titles, abbreviations, initials, examination numbers, use of dots or not in social security numbers, patient IDs, etc. The differences are generally easy to solve for a human, but it could sometimes require additional information to decide which version is correct. Medical data contained differences only in 2 % of duplicates, for example, “cancer du colon” (colon cancer) was changed to “Adénocarcinome du côlon” (colon adenocarcinoma). Some 6 % of documents were evaluated as not near-duplicates: all of these were standard forms containing only patient’s name and basic administrative data. Finally, the algorithm was not able to highlight any differences in 9 % of documents due to technical issues in PDF files (Table 3).

4. Discussion

We developed a method for detecting full- and near-duplicate medical documents and estimated that up to 5.7 % of documents received between 2020 and 2022 by the Vaud Cancer Registry had one or more full- or near-duplicates. This percentage is probably higher since the false negative rate of the Simhash method was 8 % in manually evaluated Data Set 1. This means that at least 4400 documents annually could be spared from manual and error-prone processing.

We observed that straightforward fingerprint computation on the file level was not sufficient, because the same textual content and formatting

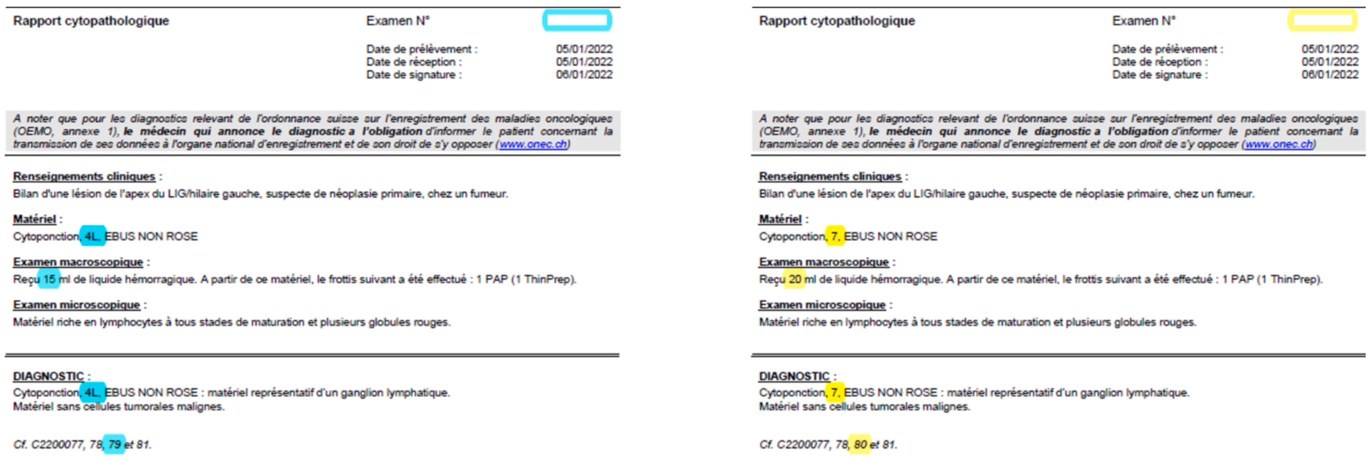

Fig. 4. Example of highlighted differences in a near-duplicate document pair.

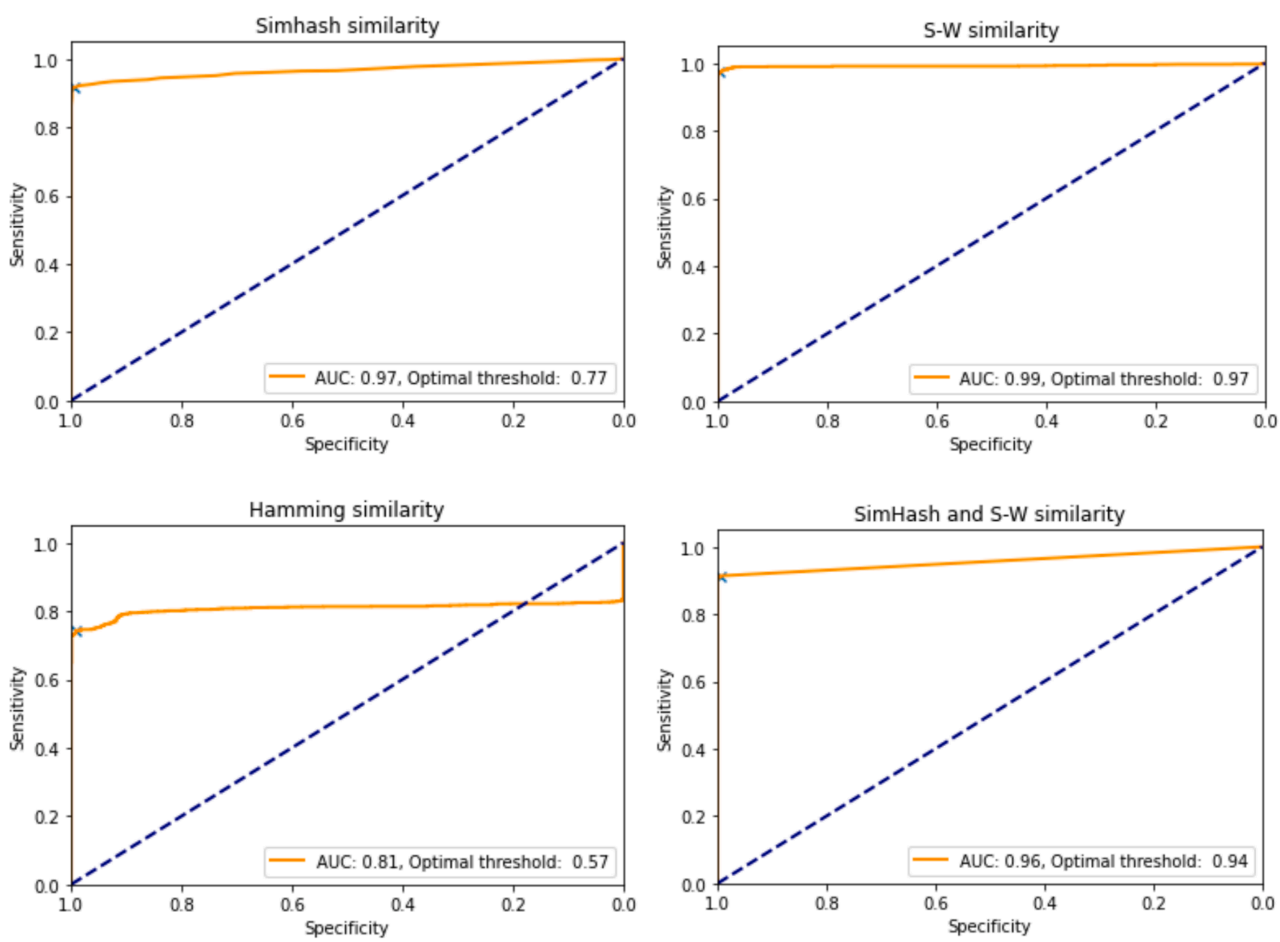

Fig. 5. ROC curves of four similarity measures against human gold standard (Data Set 1).

can be expressed in PDF documents in different ways. This can happen when PDF documents are created separately by storing different metadata (date of creation, software, font type and size, etc.). For this reason, the text content must be first extracted and preprocessed. However, some documents were near-duplicates which cannot be detected by using normal hash fingerprints. Therefore, we applied a Locality-sensitive hashing (Simhash) algorithm combined with the Smith-Waterman similarity metrics.

Full-duplicate documents can be usually processed automatically, for example by removing the new duplicate one. Near-duplicate documents are more complex to handle. No simple rule allows to automatically decide whether their information content is equal, and if not, which one of them or even both should be kept. This decision requires a human intervention. To facilitate this task, we have proposed a tool highlighting the differences. Based on our manual evaluation, 6 % of near-duplicate candidates were evaluated as not near-duplicates since differences between them, although small, were still significant (p.ex. a slightly different name and/or social security number in a standard form). Such differences are challenging for a purely syntactic method.

Our method is efficient. The total time needed for processing about

Table 2
Full- and near-duplicate content documents detection among 224,398 documents routinelycollected by the Vaud Cancer Registry (Data Set 3).

Document type	Documents (N)	%
All documents	224,398	100
Full-duplicates (same text after preprocessing)	12,256	5.46
Simhash near-duplicates	762*	0.34
Near-duplicates:		
Threshold 0.94	540	0.24
Threshold 0.97	479	0.21
Threshold 0.99	382	0.17
Full-duplicates, same patient	9,368	4.17
Full-duplicates, same transmitter	10,224	4.56
Near-duplicates (Th 0.94), same patient	298	0.13
Near-duplicates (Th 0.94), same transmitter	500	0.22

* Unit = words, width = 6, overlap = 3, k = 3 (similarity level).

Table 3
Results of manual evaluation of 298 near-duplicates.

Near-duplicate group		**Duplicate pairs (N)**	**%**
Medical data (diagnosis and measurements)		6	2 %
Administrative data	Patient-level*	63	21 %
	Doctor or clinical level*	33	11 %
	Other**	151	51 %
No visible difference		26	9 %
Not a near-duplicate		19	6 %
Total		298	100 %

*) name, address, title, contact information.
**) e.g. letter dates, 'internal copy', archive numbers.

225,000 documents was 941 s, excluding time to extract text from PDFs (network disk, Intel i-9–9900, 64 GB). Although our implementation was not optimised for speed, our method (228.2 docs/sec) has sufficient performance compared to the method presented by Gabriel et al. [4] (11.7 docs/sec) (Appendix D).

Our study has several limitations. The documents used in developing and testing stem from a single cancer registry in Switzerland, albeit a large one with multiple data producers. The current method focuses only on text and is purely syntactical. Words are treated as sequences of characters without any further meaning and all syntactical differences are given the same weight. Although the method itself is not language dependent, we tested it only for documents written in French. The French language has particularities such as commonly used accents and apostrophes which can increase the number of near-duplicates, especially in scanned documents, compared to other languages. We do not believe that there are specific limitations for using this method for other languages. Finally, in validation, we were limited to a relatively small human validated data set and simulated data.

Although we developed the method for our cancer registry, it can be easily adopted by other population-based cancer registries since most collect multisourced documents. Near-duplicate detection would also be useful in electronic health record systems to avoid storing same documents multiple times. Further, potential applications are not limited to healthcare.

Our method could be further developed in several ways. For instance, it could also determine the relevance and find the type of the difference between near-duplicate documents. Research in this direction has suggested an approach (Roul, Mittal [16] that combines MinHash and SimHash with Latent Semantic Indexing [17]. Analysing the document structure or identifying templates [18] could be applied for this purpose too. Finally, TF-IDF (term frequency–inverse document frequency) or external medical dictionaries or corpuses (e.g. [19]) could be used to give weights for terms in the Simhash algorithm to differentiate small syntactical differences from semantically meaningful ones. However, measuring semantic similarity can be challenging in medical context since small differences in the text can mean a different diagnosis or treatment regimen.

5. Conclusions

We described a method detecting duplicates and near-duplicates in cancer registry documents, validated the method against human verified data, and applied it to 225 000 medical documents. The method found that 5.5 % of documents had full-duplicates and 0.24 % near-duplicates. Some 94 % of near-duplicates were correctly detected upon manual validation while 6 % were classified as different documents.

Our implementation was based on combining the Simhash locality sensitive hashing algorithm with the Smith-Waterman text alignment similarity metric. This makes the method efficient for large document sets since only a small number of documents requires slow pairwise comparison. The method can be further improved by including semantic aspects such as weighing terms or words based on their significance and classifying differences in documents based on their type. Although the method was developed for a cancer registry, it can be applied to any medical setting where textual documents are used and (near-) duplicates exist.

The method has been implemented in the daily document processing of the Vaud Cancer Registry. It reduces the workload of medical coders. The provided visualisation tool to handle near-duplicates has been found convenient and useful.

6. Summary points

- Duplicate and near-duplicate medical documents are problematic in document management, clinical use, and medical research.
- Locality Sensitive Hashing (LSH) methods have been developed for Internet search engines and pairwise similarity metrics for text processing and biogenetics.
- This paper describes how normal and LSH hash functions (Simhash) can be combined with Smith-Waterman text alignment algorithm into a scalable full- and near-duplicate detection method.
- Differences in near-duplicate documents are highlighted to the reader.
- The method was manually verified and tested using over 224 000 multisourced cancer registry documents. Up to 6 % of documents had full- or near-duplicates.

We have replaced Patrick Arveux by Gautier Defossez in the list above as well as in the manuscript as explained in the included Authorship transfer form. If this change will not be accepted Patrick Arveux should be added to the list instead of Gautier Defossez. Their roles are the same.

CRediT authorship contribution statement

Tapio Niemi: Conceptualization, Formal analysis, Investigation, Methodology, Software, Writing – original draft. **Jean Pierre Ghobril:** Conceptualization, Investigation, Methodology, Software, Writing – original draft. **Gautier Defossez:** Resources, Validation, Writing – review & editing. **Simon Germann:** Data curation, Writing – review & editing. **Eloïse Martin:** Data curation, Software, Validation, Writing – review & editing. **Jean-Luc Bulliard:** Conceptualization, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Appendix A:. Data set descriptions

Data Set 1, which represented human interpretation of duplicates, contained 1252 duplicate and 398 near-duplicate documents. The data set was constructed as follows: we first selected a set of documents that contained many duplicates. These documents were pairwise compared using the cosine similarity metric (https://spacy.io). We then selected the most similar for a manual evaluation. The aim of the manual evaluation was determining whether document pairs were duplicates or near-duplicates, as well as writing down the noted differences. Attention was paid to medical and administrative data while differences in formatting were ignored. Finally, we added non-similar documents into the test set to get a set containing about 50 % of full- or near-duplicates.

Data Set 2 was generated by first randomly selecting 3000 non-duplicate medical documents and then adding an artificially created near-duplicate document for each original document by arbitrarily changing and removing words. This method was chosen based on manual evaluation on differences in near-duplicate documents. The arbitrary changes contained a small random number of operations in the following three groups: changing zero to two dates, mixing zero to two letters in a word, and removing zero to one word. Finally, 3042 document pairs were randomly selected from these 6000 documents to keep the same size as for Data Set 1.

Data Set 3 contained 224 398 readable PDF documents. All text of PDF documents were extracted. The minimum text length in the documents was 10 characters, maximum 265 992, mean 4301, standard deviation 4281, and median 3117.

Appendix B:. Similarity measures

We applied following similarity measures between documents A and B in the document set D as follows:

- Cosine similarity: S_C (A, B) = $\frac{A \bullet B}{\|A\|\|B\|}$, where A and B are vectors representing two documents in D.
- Normalised Simhash similarity: Ssim (A, B) = 1 – Simhash distance (A, B) / max(Simhash distance(X, Y), where X and Y are any two documents in D.
- Normalised Hamming similarity: NHS = Hsim(A, B) / max(Hsim(X, Y), where X and Y are any two documents in D, and Hsim = (A, B) = (Hamming distance (A, B) / (length(A) + length(B)).
- S-W similarity: SWsim (A, B) = number of matches / (number of matches + number of mismatches).

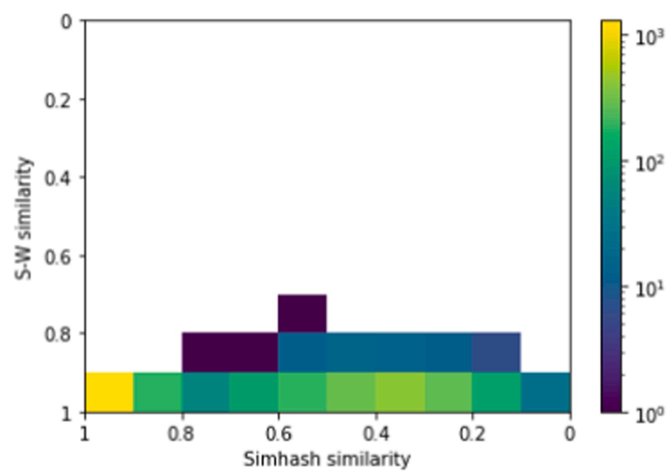

Fig. B-1. Comparison of Simhash and S-W similarity in Data set 1. The colours represent the number of document pairs in logarithmic scale (width = 6 words, overlap = 3)

Appendix C:. Algorithm for near-duplicate detection

We applied the following algorithm, in which k is the maximum number of different bits in Simhash fingerprints and th is the selected threshold for S-W similarity. We used the following values: k = 3 [6] and th = 0.94 (the optimal threshold obtained from Data set 1).

1. Read extracted text from PDF documents. (If a document is in another format, it is first converted to PDF).
2. Pre-process the extracted text. (this entails removal of extra spaces, new lines, bar codes, and non-alphanumeric special characters)
3. Compute fingerprints for each document by using a traditional hash function.
4. Compute Simhash fingerprints for each document in a multi-step process, as explained below.
5. Find near-duplicates based on Simhash fingerprints by applying an index structure and a suitable value for k.
6. Store IDs of duplicates and near-duplicate candidates as lists for each document. If the list is empty, the document has no duplicates.
7. Compute S-W similarities for each document and its near-duplicates stored in the near duplicate list. Apply a predefined threshold value, th, for the S-W similarity to select the final near-duplicate documents.
8. Additionally, classify the near-duplicates into different groups using extracted metadata such as the sender of the document or the name of the patient.
9. Detect the differences between the near-duplicates using the output of the S-W algorithm or a separate difference detection function and illustrate them (e.g., using a textual presentation or highlighting in PDF documents).

Text pre-processing was implemented based on regular expressions. No Natural Language Processing (NLP) tools were used since we are interested in all syntactical differences. Pre-processing enabled us to remove multiple white space, new lines, bar codes, and non-alphanumeric special characters. Removing full-duplicates based on hash fingerprints before computing Simhash fingerprints was not implemented. This additional step could improve performance and avoid producing 'duplicate' near-duplicate pairs.

The Simhash method applied in Steps 4 and 5 contains several steps:

1. Divide the document into suitable shingles (words, or n-grams. We use the term shingles to indicate that they can overlap). The length and overlapping of shingles affect how different near-duplicates can be. If there is no overlap, the order of the shingles in the document does not count.
2. Compute Simhash fingerprints:
   Compute hash values for each shingle using a hash function. We used Python's hashlib.md5 function to produce 64-bit long fingerprints.
   Compute the Simhash value for each shingle fingerprint as follows:
   - For each bit position, compute the sum of 1 bit and subtract the number of 0 bits.
   - The final Simhash fingerprint has 0 for every negative position and 1 for every non-negative position.
3. Use an index structure to find Simhash fingerprints differing only in k bits. These fingerprints represent near-duplicate documents.

Simhash and S-W implementations were based on https://pypi.org/project/simhash and https://pypi.org/project/swalign, respectively. The S-W implementation was modified to accept words instead of characters.

Appendix D:. Performance metrics in Data set 3 (224 398 documents)

Computing normal hash fingerprints took 2 min 12 s and Simhash fingerprints 2 min 32 s. Finding exact duplicates with hash fingerprints took 43 s while finding near-duplicate documents using the Simhash fingerprints and an index structure took 45 s. Computing S-W distances for 762 Simhash near-duplicate candidates took 9 min 19 s by using a Python function.

References

[1] J.-L. Bulliard, et al., *Nouvelles exigences légales dans le domaine de l'enregistrement du cancer: opportunités et défis [New law for cancer registration in Switzerland: opportunities and challenges]*, Rev Med Suisse 82 (2020) 2099–2103.

[2] M.D. Wang, R. Khanna, N. Najafi, *Characterizing the source of text in electronic health record progress notes*, JAMA Intern. Med. 177 (8) (2017) 1212.

[3] J.O. Wrenn, et al., *Quantifying clinical narrative redundancy in an electronic health record*, J. Am. Med. Inform. Assoc. 17 (1) (2010) 49–53.

[4] R.A. Gabriel, et al., *Identifying and characterizing highly similar notes in big clinical note datasets*, J. Biomed. Inform. 82 (2018) 63–69.

[5] A.Z. Broder, et al., *Syntactic clustering of the web*, Comput. Network. ISDN Syst. 29 (8–13) (1997) 1157–1166.

[6] G.S. Manku, A. Jain, A. Das Sarma, *Detecting near-duplicates for web crawling*, ACM Press, 2007.

[7] T.C. Hoad, J. Zobel, *Methods for identifying versioned and plagiarized documents*, J. Am. Soc. Inf. Sci. Technol. 54 (3) (2003) 203–215.

[8] Su, Z., et al. Plagiarism detection using the Levenshtein distance and Smith-Waterman algorithm. in 3rd International Conference on Innovative Computing Information and Control. 2008. IEEE.

[9] A.Z. Broder, *Identifying and filtering near-duplicate documents. Annual Symposium on Combinatorial Pattern Matching*, Springer, 2000.

[10] M.S. Charikar, *Similarity estimation techniques from rounding algorithms*, ACM Press, 2002.

[11] Henzinger, M. *Finding near-duplicate web pages*. 2006. ACM Press.

[12] Levenshtein, V.I. *Binary codes capable of correcting deletions, insertions, and reversals*. in *Soviet physics doklady*. 1966. Soviet Union.

[13] P. Jaccard, The distribution of the Flora in the alpine zone.1, New Phytol. 11 (2) (1912) 37–50.

[14] T.F. Smith, M.S. Waterman, *Identification of common molecular subsequences*, J. Mol. Biol. 147 (1) (1981) 195–197.

[15] R.W. Hamming, *Error detecting and error correcting codes*, Bell Syst. Tech. J. 29 (2) (1950) 147–160.

[16] R.K. Roul, S. Mittal, P. Joshi, *Efficient approach for near duplicate document detection using textual and conceptual based techniques*, in: *Advanced Computing, Networking and Informatics-Volume 1*, Springer, 2014, pp. 195–203.

[17] Deerwester, S., Improving information retrieval with latent semantic indexing, in Proceedings of the 51st Annual Meeting of the American Society for Information Science 25. 1988. pp. 36-40.

[18] A.M. Redd, et al., *A pilot study of a heuristic algorithm for novel template identification from VA electronic medical record text*, J. Biomed. Inform. 71 (2017) S68–S76.

[19] N. Grabar, C. Dalloux, V. Claveau, *CAS: corpus of clinical cases in French*, J. Biomed. Semant. 11 (1) (2020).